



Optimizing financial decision-making for emerging adults: A compact Python-based personalized financial projection approach

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ABSTRACT

The financial decisions made by emerging adults significantly influence their long-term financial well-being. To make effective decisions, they need both financial literacy and a forward-looking mindset. However, existing literature lacks tailored financial interventions for this demographic. Addressing this gap, our study implemented and evaluated a Python-based Personalized Financial Projection (PFP) among emerging adults in Hong Kong. Our findings indicate that the Python-based PFP effectively promoted prudent financial behaviors by enhancing participants' perceived financial literacy and future time perspectives. This represents a significant advancement, providing the first empirical evidence supporting the integration of computer programming into financial education initiatives. In an era of interactive Artificial Intelligence (AI) for financial guidance, the Python-based PFP stands out as a pivotal resource capable of leveraging interactive AI to enhance financial decision-making processes.

1. Introduction

Making intelligent financial decisions during emerging adulthood is crucial, as these choices significantly influence an individual's long-term financial well-being. Unlike adolescents, emerging adults have greater control over their finances. However, irresponsible credit behaviors, such as late payment of credit card bills, can harm their credit records and negatively impact long-term financial health [1–3]. Additionally, the decision to enter the labor market or pursue higher education after high school graduation leads emerging adults onto different life paths. Making correct financial decisions during this period contributes to sustainable personal wealth growth throughout life [4–6].

Given the importance of promoting future-oriented financial behaviors and avoiding shortsighted decisions during emerging adulthood, financial education targeting this age group should emphasize the power of compound interest—a vital component of financial literacy that illustrates the future value and cost of money—and foster a future time perspective (optimistic outlook on the future) to encourage forward-thinking financial decisions [7,8]. However, a comprehensive review reveals that there are significantly fewer validated financial education programs designed specifically for emerging adults compared to those for children and adolescents, especially programs that address both financial literacy and future time perspectives [9].

Among various financial education approaches, personalized financial projection (PFP) shows promise in promoting future time perspective and understanding of compound interest [10–14]. Typically, PFPs involve a mobile application where participants input parameters such as current income, present wealth, saving behaviors, and investment portfolio. The application then projects the total wealth participants can expect to accumulate in the future based on their inputs.

1.1. Previous research of our team

Our team recently introduced PFP to middle-aged working adults aged 36–55 in Hong Kong and conducted randomized controlled trials, finding that PFP significantly promoted future time perspectives and financial planning in the short, medium, and long terms [15]. Post-hoc analysis, dividing the sample into younger and older subsamples, revealed more pronounced effects among the younger subsample, suggesting potentially increased effectiveness of PFP among young working adults. Additionally, we noted that PFP implemented through a mobile app has limitations in promoting financial literacy since users only see the projection outcomes without understanding the underlying calculations. Furthermore, a mobile app-based PFP has limitations in strongly affecting participants' future time perspectives because it does not allow for model adjustments based on specific participant situations.

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To address these limitations—participant age and the mobile app’s deficiencies—we employed a Python-based PFP in a subsequent experiment, targeting young working adults aged 18–35 [16]. Python serves as a user-friendly programming tool, enabling participants to understand the PFP mechanisms at a low cost [17]. Thus, Python was used to familiarize young working adults with the coding behind PFP [16]. Our findings revealed that this initial Python-based PFP significantly improved financial planning, but the impact was modest and not mediated by enhanced financial literacy or heightened future time perspectives [16].

Participant feedback highlighted several factors limiting the initial version’s efficacy. Firstly, participants over 30 found Python coding more challenging. Secondly, the intervention’s duration (7 h) was excessive, leading to fatigue, with 5 h dedicated to basic Python coding and only 2 h to projection exercises. Thirdly, the projection models were overly generalized and did not adequately address specific financial decisions relevant to young adults, such as postgraduate education investments and housing loans.

1.2. This current research

In response, this study developed a concise and refined Python-based PFP, reducing the duration to 120 min, with only 45 min for basic Python coding instruction, allowing more time for Python-based PFP tasks. This version features themed projection models providing advice on key financial decisions for emerging adults, such as housing loans and tuition fees for higher education. We specifically limited participants to college students aged 18–30, ensuring the inclusion of young digital natives who found learning Python coding relatively easy. The primary aim of this research was to conduct a randomized controlled trial to determine whether the concise Python-based PFP significantly enhanced financial literacy, future time perspective, and forward-thinking financial decision-making.

2. Literature review

2.1. Background

2.1.1. Importance of Python-based PFP in the era of interactive artificial intelligence

The rapid evolution and refinement of interactive artificial intelligence (Interactive AI) models empower users to project their financial trajectories and make real-time financial decisions informed by readily available information garnered through dialogue. Nonetheless, the inherent opacity of these complex models, characterized by their “black box” nature, impedes users from delving into the technical intricacies and detailed processes underpinning the projections [18,19]. This lack of transparency may erode “trust” in the models for several reasons. Firstly, users may harbor concerns about the algorithm’s ability to accurately encapsulate various economic life events in a manner deemed equitable [20]. Secondly, the majority of users lack proficiency in computer programming and the fundamental tenets of embedded mathematical models, rendering it arduous for them to bridge the gap between the projection outcomes generated by interactive AI and their practical implications in daily life [21]. Consequently, the resultant deficit in trust may hinder the efficacy of AI-driven messages in shaping users’ financial attitudes, decisions and behaviors.

To redress this lacuna, recent scholarship advocates for developers to elucidate the inner workings of projection models to engender trust among users [20]. However, extant proposed remedies, such as visualizing underlying mechanisms and deploying attention-based explanations, cater more to users already versed in the intricacies of projection models, rather than lay users seeking applicability to their own contexts [22,23]. In a novel approach, recent studies posit that employing a “counterfactual” design proves most efficacious in elucidating projection models to users, as it aligns with the logical reasoning inherent in

human cognition [24]. When contemplating an uncertain future, individuals instinctively ponder the ramifications of altering initial conditions on the ensuing outcomes [25]. Thus, the adoption of counterfactual design holds promise in bolstering users’ trust in projection outputs and effecting genuine changes in their financial attitudes and decision-making.

Our Python-based PFP embodies a counterfactual framework, as we furnish users with the requisite coding, empowering them to comprehend and fine-tune projection parameters, thereby yielding a myriad of counterfactual results. Consequently, if the Python-based PFP proves efficacious, it heralds a novel avenue for enhancing trust in interactive AI and shaping financial dispositions and behaviors. In future projections, users may opt to solicit interactive AI to generate intelligible and potent financial projection “codes”, instead of merely requesting projection outcomes. It is widely acknowledged that interactive AI excels in crafting highly structured projection codes. For users, the unveiling of the black box underlying the projection model imbues the projection process with greater credibility and persuasiveness, with the salient limitations of interactive AI poised to be rectified.

2.1.2. Importance of introducing Python-based PFP to emerging adults in Hong Kong

Recent evidence highlights the unpreparedness of emerging adults in Hong Kong to face the challenges of the post-pandemic economic landscape. Their financial behaviors often lack a future-oriented perspective, and their decision-making skills in managing finances are insufficient, potentially leading to a decline in their financial well-being. A report by the Investor and Financial Education Council (IFEC) in March 2020 reveals that only about half of young working adults in Hong Kong set financial goals and prepare for financial emergencies, compared to 82 % of mature working adults. The report also indicates that 35 % of young people were more likely to overspend, exceeding their financial means, compared to approximately 20 % of the general population [26]. Furthermore, 32 % of young individuals were burdened by debt, whereas only 12 % of the general respondents reported the same [26]. In terms of financial services, young people exhibited riskier behavior, with 27 % admitting to making only minimum payments on their credit card bills, compared to just 7 % of the general population [26]. Consequently, only 23 % of young working adults expressed satisfaction with their financial status, in contrast to 39 % of all respondents. These statistics have raised concerns among educators, emphasizing the need for financial education to enhance financial literacy, future time perspectives, and healthy financial practices among emerging adults.

However, the latest systematic review suggests that most validated financial education programs internationally are standardized courses primarily targeting children and adolescents, while the number of validated financial education programs for emerging adults remains limited, with mixed effects [9]. Unlike adolescents, implementing full-length standardized financial education programs among emerging adults is challenging due to their demanding schedules. Research indicates that financial interventions during emerging adulthood should be topic-specific and have clear objectives aimed at enhancing specific aspects of financial well-being. In this context, Python-based PFP, with its clear objectives of promoting an understanding of compound interest, fostering future time perspectives, and motivating future-oriented financial behaviors, appears to be a timely approach that financial educators and practitioners in Hong Kong should consider.

2.2. Intervention mechanism of Python-based PFP

2.2.1. Effects of Python-based PFP on future time perspectives

People typically do not actively think about or invest in the future when they perceive it as uncertain [27,28]. When individuals believe the future is filled with uncertainties, they tend to disconnect current decisions from future outcomes. This psychological disconnection, as

conceptualized by Bartels and Urminsky [29], raises concerns among behavioral scientists. Such disconnection reduces the likelihood of engaging in future-oriented behaviors, including physical exercise, maintaining a healthy diet, and practicing sound financial behaviors such as regular saving, budgeting, spending control, and avoiding debt [30].

Previous initiatives have employed various technologies to establish a link between an individual's present and future selves [31–33]. These initiatives aim to enhance participants' future time perspectives by making the future more vivid based on present information. For example, age-progressed renderings [33,34] generate a visual representation of the future self by applying changes in facial parameters that typically occur during the aging process (e.g., skin laxity). These renderings serve as a bridge between the present and future images. Similarly, AI can create a future-self based on current parameters and enable communication between the current and future selves [31,32]. Additionally, PFPs have gained popularity as a means of forecasting users' future financial status by incorporating their current financial behaviors and simulating market fluctuations [10–14]. Empirical findings strongly support the effectiveness of age-progressed renderings, future-self communication, and PFPs in promoting future time perspectives [13, 14,31,32]. Given that the Python-based PFP is essentially a form of PFP, it incorporates all the features associated with PFPs and is expected to cultivate an optimistic outlook towards the future.

2.2.2. Effects of Python-based PFP on financial literacy

The widely accepted “big-three” measurement of financial literacy emphasizes the importance of understanding compound interest, inflation, and risk diversification [35]. In this study, one objective of implementing Python-based PFP was to enhance the understanding of compound interest component of financial literacy. Krishnamurthy and Sujan [36] highlighted a common issue in compound interest: Individuals often fail to grasp how savings and debts grow exponentially through compound interest provided by commercial banks [36]. Similarly, McKenzie and Liersch [37] discovered that people tend to consistently underestimate the exponential growth of savings and debts due to their limited financial literacy. One possible reason is that the human brain struggles to calculate comprehensively the long-term effects of compound interest on the accumulation of wealth or debt [29]. In this context, the advantages of Python-based PFP in promoting financial literacy become evident.

First, participants are guided to input the correct compound interest formula, enabling them to simulate the exponential growth of personal savings and debts. Through multiple simulation rounds, participants can observe significant and “surprising” growth of their savings or debts, thereby gaining a firsthand understanding of the power of compound interest. Second, Python-based PFP can provide participants with the accumulated savings or debts at the end of each simulation round, facilitating their comprehension of how compound interest drives exponential growth over time. Third, participants have the opportunity to adjust critical parameters, such as increasing their monthly savings, allowing them to witness how their choices can further amplify the impact of compound interest. Considering these advantages, we believe that Python-based PFP has the potential to significantly enhance participants' financial literacy. By actively engaging with the simulation, participants can develop a deeper understanding of compound interest and its profound influence on their financial well-being.

2.2.3. Hypothetical model

Based on the theoretical discussion above, this study aimed to investigate the impact of Python-based PFP on financial literacy, future time perspectives, and financial behaviors among emerging adults, while considering the mediating role of financial literacy and future time perspectives. These three outcome variables were of utmost importance for our evaluation. Given the absence of a formally validated assessment tool for financial literacy among emerging adults in Hong Kong, we

assessed financial literacy through subjective financial understanding (perceived financial literacy) rather than objective financial knowledge. Due to the limited duration of the intervention project, we were only able to measure the immediate effects of Python-based PFP and could not track the medium- and long-term effects. Since immediate changes in financial behaviors may not be readily detectable, we utilized three instrumental variables as substitutes for financial behaviors: financial behavioral control, tendency to engage in healthy financial behaviors, and life satisfaction. Previous studies conducted in Hong Kong and other economies have demonstrated significant and positive correlations between these constructs and financial behaviors among emerging adults [38–40]. By incorporating these variables, we strengthened the robustness of our findings. Fig. 1 illustrates the hypothetical relationships between Python-based PFP and the five outcome constructs, based on the established pathways in previous investigations [38–40]. Upon reviewing the hierarchical structure depicted in Fig. 1, the primary focus of this research was to examine the effectiveness and mechanism of Python-based PFP. Specifically, our hypotheses were as follows:

- Python-based PFP improves perceived financial literacy, future time perspectives, financial behavioral control, tendency to engage in healthy financial behaviors, and life satisfaction, respectively (Hypothesis 1);
- Python-based PFP enhances the tendency to engage in healthy financial behaviors through the mediating effects of perceived financial literacy, future time perspectives, and financial behavioral control (Hypothesis 2);
- Python-based PFP promotes life satisfaction by mediating an increased tendency to engage in healthy financial behaviors (Hypothesis 3).

3. Method

3.1. General design

To ensure the suitability of participants for the Python-based PFP targeting emerging adults in Hong Kong, we established specific recruitment criteria: 1) Participants had to be between 18 and 30 years of age, 2) They should not be majoring in finance or computer science, 3) They should have no prior completion of standardized financial courses, and 4) They should have no proficiency in Python programming. Recruitment was restricted to a humanities and social sciences public university in Hong Kong, based on the likelihood that their full-time students would meet these criteria.

Among the participants, we assessed five outcome variables: perceived financial literacy, future time perspectives, financial behavioral control, tendency to perform healthy financial behaviors, and life satisfaction. To evaluate the effects of the Python-based PFP on these constructs, we conducted a two-arm randomized control trial, randomly assigning participants to either the experimental or control group. All participants were invited to participate in the assessment of outcome variables at two time points: pretest and posttest.

3.2. Procedures

The Registry section of the university sent mass invitations via bulk email to full-time undergraduate and postgraduate students, aiming to recruit participants for our research. Participants who completed the pretest, posttest, and intervention received a Cash Coupon worth HK \$100. We successfully recruited 78 participants from the campus community, randomly assigning 40 to the experimental group and 38 to the control group.

Participants in the experimental group could choose one of three intervention sessions, each focusing on Python-based PFP. Assignment to a specific session was based on participants' preferences, with an effort to ensure a balanced distribution of approximately 13–15

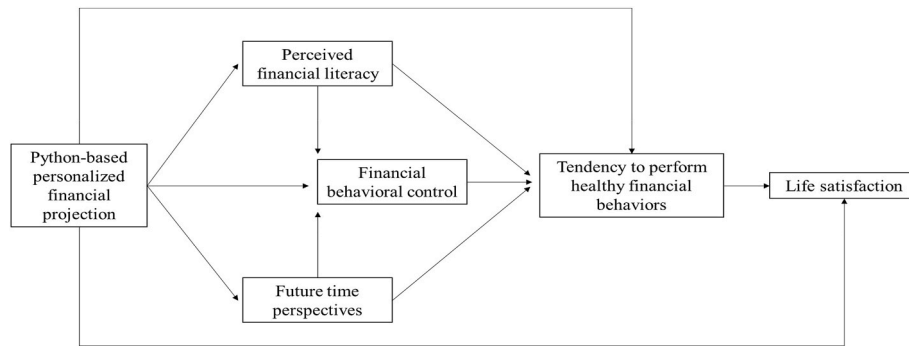


Fig. 1. Hypothesized working mechanism of Python-based personalized financial projection.

participants per session. Students participated in a baseline assessment before the training and a posttest after the training to eliminate potential attrition bias. They were informed during the baseline test that they could withdraw from the research at any stage without providing a reason.

The three sessions of Python-based PFP training took place on 27th February, 1st March, and March 2, 2023, in computer labs, with each participant having access to an individual computer. Meanwhile, the 38 students assigned to the control group attended an unrelated statistics course on 29th March, providing a contrasting experience. Ethical approval was obtained from the Research Committee of Lingnan University (Hong Kong) for conducting the data collection, underscoring our commitment to upholding research ethics and protocols.

3.3. Measures

Perceived financial literacy was assessed by asking students to rate their understanding of financial knowledge on a seven-point scale, ranging from 1 (lowest level) to 7 (highest level). We utilized six questions adapted from Tomar et al. [41], originally developed by Koposko and Hershey [42], to measure future time perspectives. Example items included statements such as “I look forward to life in the distant future” and “My friends describe me as future-oriented.” These items were adjusted to a five-point Likert scale, where 1 represented “strongly disagree” and 5 represented “strongly agree.” The Cronbach’s Alpha reliability coefficients for all six items were 0.88 for the pretest and 0.91 for the posttest. The mean score of the six items was calculated to determine participants’ future time perspectives.

Financial behavioral control was assessed using one item from Xiao et al. [43] and three items developed by our research team. Participants rated how easy or difficult it was for them to stick to their financial plans on a five-point scale, ranging from 1 (very difficult) to 5 (very easy). The internal consistencies for the pretest and posttest scores were 0.91 and 0.93, respectively. The mean score of the four items was computed to represent financial behavioral control.

To evaluate the tendency to perform healthy financial behaviors, participants indicated the extent to which they engaged in seven positive financial behaviors on a five-point scale ranging from 1 (strongly disagree) to 5 (strongly agree). The behaviors included saving regularly, tracking monthly expenses, spending within a budget, maintaining an adequate balance in their bank account, saving for emergencies, saving for the future, and investing regularly. These items were adapted from Shim et al. [38] and Xiao et al. [44]. The internal consistencies for the pretest and posttest scores were 0.81 and 0.85, respectively. The mean score of the seven items was calculated to determine participants’ tendencies toward healthy financial behaviors.

Life satisfaction was measured using five items validated among college students in Hong Kong by Sachs [45]. Sample items include “In most ways, my life is close to my ideal” and “So far, I have achieved the important things I want in life.” Responses were calibrated on a

five-point scale, ranging from 1 (strongly disagree) to 5 (strongly agree). The internal consistencies for the pretest and posttest scores were 0.87 and 0.91, respectively. The mean score of the five items was calculated to reflect participants’ life satisfaction.

3.4. Design of Python-based PFP

Participants underwent a comprehensive 45-min training session, where they received instruction in fundamental coding skills and the grammatical rules of Python. Following this, they participated in a 75-min session involving Python-based personalized financial projections. Our research team developed three distinct financial projection models using Python, focusing on critical financial decisions during emerging adulthood: housing mortgages, education tuition loans, and regular savings. These models aimed to simulate the impact of these decisions on participants’ financial well-being in both the near and distant future.

Equipped with basic Python coding skills, participants were encouraged to modify key parameters within the models to observe the far-reaching consequences of their current debt and saving behaviors. The housing mortgage projection model provided insights into the true cost of owning an apartment throughout one’s life, considering factors such as interest payments to a commercial bank. By working with this model, participants could see how an imprudent decision to buy a house could negatively impact future cash flows, delaying their attainment of financial independence.

The tuition loan projection model presented the actual cost of investing in higher education degrees, incorporating interest expenses. This model enabled students to make economically prudent choices aligned with their families’ financial circumstances and their anticipated future income. Additionally, students gained insights into the potential repercussions of selecting inappropriate degree programs, which could lead to significant debt burdens.

The savings projection model emphasized the importance of early engagement in regular savings, highlighting how it can lead to greater wealth accumulation in the future. Collectively, these Python-based PFP models provided participants with practical knowledge and tools to enhance their understanding of the long-term impacts of their present financial decisions. Consequently, participants were empowered to make more informed choices and secure their financial well-being in the future.

3.5. Data analysis

To ensure the validity of the group assignments (control group vs. experimental group), we conducted a comprehensive set of statistical tests to assess any significant differences between the two groups in terms of outcome and background variables. Specifically, we employed independent t-tests to compare the means of continuous variables across the two groups, including age, monthly household income, household size, perceived financial literacy, future time perspectives, financial

behavioral control, tendency to perform healthy financial behaviors, and life satisfaction. Additionally, we used the chi-square test to examine gender proportion differences between the experimental and control groups.

For our principal analysis, we utilized multiple regression analysis to determine whether participants in the experimental group reported higher levels of perceived financial literacy, greater future time perspectives, increased financial behavioral control, a stronger tendency to engage in healthy financial behaviors, and higher life satisfaction compared to those in the control group. It was crucial to control for the baseline status of these key constructs to ensure robust analysis. Furthermore, we included relevant background variables in the regression model to validate the results and account for any potential confounding factors. Subsequently, we incorporated the pretest and posttest data into a structural model to evaluate the mechanism of the Python-based PFP, specifically examining how it influenced the five outcome variables in sequence.

4. Results

Table 1 presents the results of the chi-square and t-tests, indicating no significant differences across all variables, except for age (which was controlled for in the main analysis), between the experimental group ($N = 40$) and the control group ($N = 38$). This confirms the effectiveness of our group assignment. Table 2 shows that the Python-based PFP had the anticipated positive impact on all outcome variables. Initially, without controlling for background variables, the experimental condition demonstrated significant improvements in perceived financial literacy ($\beta = 0.16$, $p < 0.01$), future time perspectives ($\beta = 0.23$, $p < 0.01$), financial behavioral control ($\beta = 0.15$, $p < 0.01$), tendency to perform healthy financial behaviors ($\beta = 0.18$, $p < 0.01$), and life satisfaction ($\beta = 0.19$, $p < 0.05$). When background variables were included as controls, all effects remained significant and positive. Specifically, significant enhancements were observed in perceived financial literacy ($\beta = 0.18$, $p < 0.05$), future time perspectives ($\beta = 0.13$, $p < 0.10$), financial behavioral control ($\beta = 0.14$, $p < 0.10$), tendency to perform healthy financial behaviors ($\beta = 0.20$, $p < 0.01$), and life satisfaction ($\beta = 0.20$, $p < 0.05$). Therefore, the Hypothesis 1 is fully supported.

Fig. 2 presents the results of the structural models, illustrating the estimated coefficients of all direct and indirect paths from the intervention to the outcome variables. Three key observations emerge from the findings. Firstly, the effect of Python-based PFP on the tendency to perform healthy financial behaviors was fully mediated by increases in perceived financial literacy ($\beta = 0.17$, $p < 0.01$; $\beta = 0.13$, $p = 0.05$) and improved future time perspectives ($\beta = 0.24$, $p < 0.01$; $\beta = 0.22$, $p <$

Table 1
Descriptive statistics of the control and experimental groups.

	Control group ($N = 38$)	Experimental group ($N = 40$)
	%/mean (standard deviation)	
Male (%)	47.4	40.0
Age	26.11 (3.52) ^b	22.97 (3.41) ^b
Family income ^a (Range: 1–8)	3.33 (2.28)	3.29 (2.14)
Household size	3.08 (0.89)	3.27 (0.96)
Perceived financial literacy (Range: 1–7)	3.87 (1.30)	4.13 (1.27)
Future time perspectives (Range: 1–5)	3.93 (0.61)	4.10 (0.74)
Perceived behavioral control (Range: 1–5)	3.13 (0.90)	2.97 (0.81)
Tendency of performing healthy financial behaviors (Range: 1–5)	3.70 (0.55)	3.63 (0.64)
Life satisfaction (Range: 1–5)	3.27 (0.85)	3.14 (0.77)

Note.

^a Monthly family income ranged from 1 (HK\$18,700 – HK\$19,999) to 8 (HK\$100,000 or above).

^b $p \leq 0.01$.

Table 2
Results of experiment-based multiple linear regression models.

	Posttest score of perceived financial literacy		Posttest score of future time perspectives		Posttest score of financial behavioral control	
Joining the intervention	0.16 ^c	0.18 ^b	0.23 ^c	0.13 ^a	0.15 ^c	0.14 ^a
Baseline score of outcome variable	0.83 ^c	0.07 ^c	0.66	0.72 ^c	0.78 ^c	0.75 ^c
Being male		0.17 ^b		−0.02		0.20 ^c
Age		0.03		−0.26 ^c		−0.05
Family income		0.05		−0.10		−0.04
Household size		0.09		−0.13		0.01
	Posttest score of tendency to perform healthy financial behaviors		Posttest score of life satisfaction			
Joining the intervention	0.18 ^c	0.20 ^c	0.19 ^b	0.20 ^b		
Baseline score of outcome variable	0.82 ^c	0.85 ^c	0.68 ^c	0.68 ^c		
Being male		−0.03		0.05		
Age		0.02		0.02		
Family income		−0.03		0.03		
Household size		−0.14 ^b		0.02		

Note. $N = 78$. Standardized estimated coefficients are reported.

^a $p \leq 0.10$.

^b $p \leq 0.05$.

^c $p \leq 0.01$.

0.01), respectively, partially supporting the Hypothesis 2. Secondly, the impact of Python-based PFP on life satisfaction was partially mediated by increases in perceived financial literacy and the tendency to perform healthy financial behaviors ($\beta = 0.17$, $p < 0.01$; $\beta = 0.13$, $p = 0.05$; $\beta = 0.34$, $p < 0.01$). Additionally, the influence on life satisfaction was partially mediated by improved future time perspectives and the tendency to perform healthy financial behaviors ($\beta = 0.24$, $p < 0.01$; $\beta = 0.22$, $p < 0.01$; $\beta = 0.34$, $p < 0.01$). Thus, the Hypothesis 3 is well supported. Thirdly, the direct effect of the intervention on financial behavioral control disappeared when controlling for perceived financial literacy and future time perspectives as predictors of financial behavioral control. These findings highlight the multifaceted impact of Python-based PFP on enhancing financial literacy, future time perspectives, and financial behaviors, ultimately contributing to greater life satisfaction among emerging adults in Hong Kong.

5. Discussion

This study implemented a Python-based personalized financial projection for emerging adults in Hong Kong. The positive findings from the randomized control trial suggest that Python-based PFP is a timely and effective approach to addressing short-sighted financial decision-making among emerging adults, particularly in the post-pandemic economic context. This intervention can facilitate financial independence and secure long-term financial well-being. The study advances the theories and practice of personal finance education in four main aspects.

First, the demonstrated effectiveness of Python-based PFP may encourage future studies to further integrate “computer-assisted learning” into financial education. This research is a pioneering effort to incorporate programming into financial education on an international scale. Our promising findings may inspire other scholars to replicate this study or develop additional programming-based financial education modules. While Humber [17] developed personal finance management modules with Python and suggested their extension to young consumers, there has been no formal experiment-based assessment to validate the effects of Python-based financial modules on users’ knowledge and behaviors. Thus, this research could be the first worldwide to test the effectiveness of Python-based financial interventions through a randomized control trial.

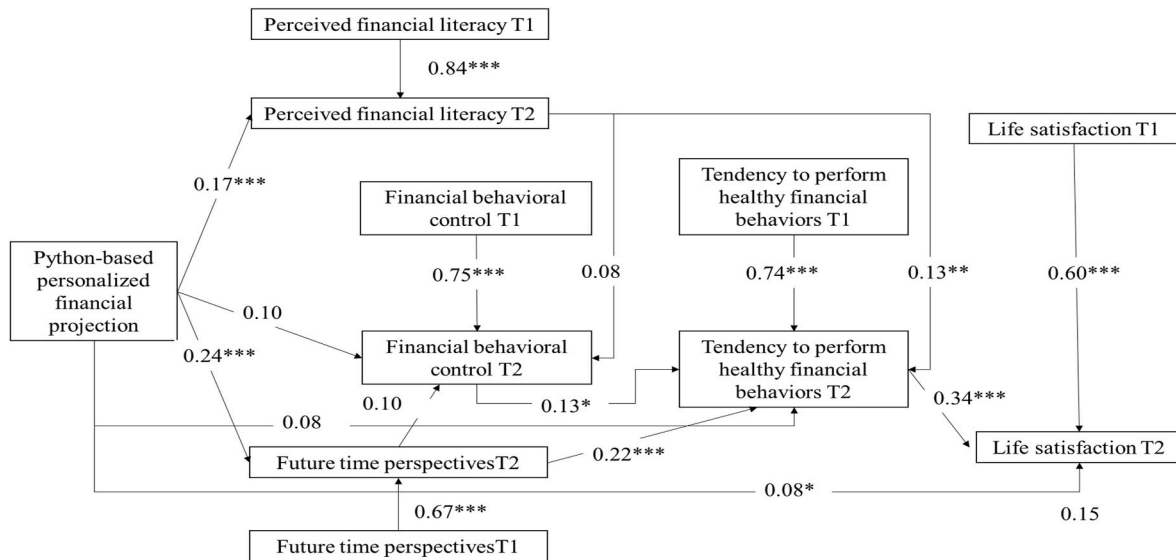


Fig. 2. Results of experiment-based structural model. Note. $N = 78$. Standardized estimated coefficients are reported. $*p \leq 0.10$; $**p \leq 0.05$; $***p \leq 0.01$.

Second, the design and evaluation of Python-based PFP significantly contribute to the development of “personalized” financial education. Personalized financial education offers a unique advantage over generic financial education by targeting underlying psychology to bring about positive behavioral changes [10–14]. Instead of simply instructing participants with commonly accepted knowledge through standardized courses, personalized financial education guides participants to make the best choices based on their unique circumstances. While interactive mobile applications have activated the development of personalized financial interventions (e.g., PFP), the combination of Python and PFP unleashes the full potential of PFP, allowing users to adjust settings and parameters according to their specific situations without any technical limitations. In other words, Python-based PFP creates personalized financial projections that are more closely aligned with reality.

Third, the development of Python-based PFP offers a promising direction for financial education targeting emerging adults. Unlike children and adolescents, who can undergo highly structured, comprehensive financial education, emerging adults often have busy schedules and can only engage in topic-based financial training [9]. Our Python-based financial training, compared to previous initiatives for young people, is more targeted and addresses a critical gap in their financial understanding, specifically the effects of compound interest. Our intervention helps emerging adults comprehend the real cost of debts and the benefits of savings over time, encouraging them to think about the future and engage in future-oriented financial behaviors. Python-based PFP can be seen as a cost-effective initiative for developing financial training for emerging adults, and its effectiveness may inspire future research to develop more topic-specific financial education for young individuals.

Fourth, the successful validation of Python-based PFP can significantly enhance the smart utilization of interactive AI for personalized financial decision-making. In an era where interactive AI has garnered widespread acceptance, users may naturally gravitate towards seeking financial projections or advice directly from such systems. However, the opaque nature of the projection process and concerns regarding algorithmic fairness (e.g., the possibility that critical economic events may not be equally accounted for throughout the projection) may lead users to harbor reservations about the accuracy and reliability of the AI-generated outcomes [18–20]. This potential lack of trust could undermine the efficacy of AI-generated outcomes in influencing users’ financial attitudes and behaviors. Our validated Python-based PFP offers a smart approach to harnessing the power of interactive AI for financial decision-making. We advocate for a methodology wherein users request

the AI to generate Python-based projection code, rather than receiving direct projection outcomes and financial recommendations. This projection code serves the dual purpose of elucidating the projection process to users and fostering their comprehension of the underlying mathematical principles and philosophies. Moreover, the provision of projection code empowers users to engage in scenario analysis, enabling them to simulate potential future scenarios by altering their current decisions. This counterfactual approach mirrors human logical reasoning and is poised to have a significant impact on altering cognitive perceptions and attitudes [24]. In essence, the Python-based PFP serves as a conduit to unlock the full potential of interactive AI in shaping financial attitudes and, ultimately, enhancing users’ decision-making capabilities.

5.1. Limitations

While our contributions to personal finance and the personal finance education framework are noteworthy, it is important to acknowledge the limitations of this research, which provide strong motivation for future studies to replicate or optimize the design. Firstly, a significant limitation of our study is the lack of a validated objective measure of financial literacy that comprehensively covers saving, budgeting, borrowing, and investment among emerging adults. It is worth noting that a considerable proportion of young people in Hong Kong tend to overestimate their financial literacy and exhibit financial overconfidence [46]. Therefore, the inclusion of an objective measure of financial literacy would provide a more accurate assessment of participants’ financial knowledge and skills, thus strengthening the overall study design.

Secondly, due to the limited duration of the project, we only examined the immediate effect of Python-based PFP. This design constraint restricted our selection of outcome variables, as we had to exclude constructs that require a longer time frame to detect changes, such as actual behavioral changes and objective measures of financial behaviors (e.g., savings outcomes, debt levels, and number of credit cards held). To gain a more comprehensive understanding of the intervention’s impact, we recommend that future research optimize the study design by incorporating multiple waves of post-intervention data collection. This would allow for the examination of the middle- and long-term effects of Python-based PFP. Additionally, the inclusion of a broader range of outcome variables would enable a more sophisticated analysis of the intervention’s working mechanisms and provide a deeper understanding of how Python-based PFP improves the financial well-being of

participants.

Thirdly, our sample consisted solely of individuals without any prior programming experience because our study targeted a broad spectrum of emerging adults. Nonetheless, it stands to reason that the Python-based PFP could hold even greater appeal among those proficient in programming. Their adeptness in coding may enable them to devise more precise projection code, potentially yielding more profound psychological shifts and fostering more positive behavioral changes. In light of this, we advocate for future research endeavors to delve into the efficacy of the Python-based PFP within a subset of the emerging adult population—specifically, those with backgrounds and expertise in computer science. By focusing on this cohort, researchers can explore how their pre-existing coding skills moderate the effects of Python-based PFP on financial literacy and future time perspectives and their subsequent impacts on financial decision-making.

6. Conclusion

In the age of interactive AI, this research introduces a timely innovation: Python-based PFP aimed at emerging adults. Through rigorous validation, this study advances financial education for this demographic and offers a cost-effective strategy for leveraging interactive AI in personal financial decision-making. The implications of this research extend beyond its immediate scope, promising transformative changes in the global financial education landscape. By harnessing Python-powered financial interventions, this study paves the way for a shift from traditional, one-size-fits-all financial education models to personalized interventions that address the psychological foundations of young individuals, influencing their financial behaviors over the long term. Encouraging users to request projection code from interactive AI, rather than direct outcomes, fosters greater trust in AI systems, thereby enhancing their appeal among emerging adults. This user recognition and endorsement are pivotal in accelerating the adoption and advancement of interactive AI technologies.

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Ethical statement

Ethical approval for this study was obtained from the Research Committee of The Lingnan University (Hong Kong) prior to data collection.

Informed consent

Informed consent was obtained from all individual participants included in the study.

CRedit authorship contribution statement

Alex Yue Feng Zhu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declares no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

Data availability

Data will be made available on request.

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